Appium 2.x Migration Map

Key Changes Required

1. Dependencies Update

• Appium Java Client: 7.3.0 → 8.6.0+
• Selenium: 3.141.59 → 4.15.0+
• Cucumber: 1.2.5 → 7.14.0+
• TestNG: 6.9.9 → 7.8.0+

2. Class Replacements

Old (Appium 1.x)	New (Appium 2.x)	Notes
`MobileElement`	`WebElement`	Universal element interface
`Android-Driver<MobileElement>`	`AndroidDriver`	No generic type needed
`IOSDriver<MobileElement>`	`IOSDriver`	No generic type needed
`AppiumDriver<MobileElement>`	`AppiumDriver`	No generic type needed
`DesiredCapabilities`	`UiAutomator2Options` / `XCUITestOptions`	Platform-specific options
`MobileBy`	`AppiumBy`	Updated locator strategies
`TouchAction`	W3C Actions API	Use PointerInput and Sequence

3. Import Changes

```
// Remove these imports
import io.appium.java_client.MobileElement;
import io.appium.java_client.TouchAction;
import org.openqa.selenium.remote.DesiredCapabilities;
import org.openqa.selenium.interactions.touch.TouchActions;

// Add these imports
import org.openqa.selenium.WebElement;
import io.appium.java_client.android.options.UiAutomator2Options;
import io.appium.java_client.ios.options.XCUITestOptions;
import io.appium.java_client.AppiumBy;
import org.openqa.selenium.interactions.PointerInput;
import org.openqa.selenium.interactions.Sequence;
```

4. Driver Initialization Changes

```java
// Old way
DesiredCapabilities capabilities = new DesiredCapabilities();
capabilities.setCapability("platformName", "Android");
driver = new AndroidDriver<MobileElement>(new URL("http://0.0.0.0:4723/wd/hub"), capab-
ilities);

// New way
UiAutomator2Options options = new UiAutomator2Options();
options.setPlatformName("Android");
driver = new AndroidDriver(new URL("http://0.0.0.0:4723/"), options);
```

5. Element Finding Changes

```java
// Old way
MobileElement element = driver.findElement(MobileBy.id("elementId"));

// New way
WebElement element = driver.findElement(AppiumBy.id("elementId"));
```

6. Touch Actions Replacement

```java
// Old TouchAction
new TouchAction(driver).tap(PointOption.point(x, y)).perform();

// New W3C Actions
PointerInput finger = new PointerInput(PointerInput.Kind.TOUCH, "finger");
Sequence tap = new Sequence(finger, 1);
tap.addAction(finger.createPointerMove(Duration.ofMillis(0), PointerInput.Origin.view-
port(), x, y));
tap.addAction(finger.createPointerDown(PointerInput.MouseButton.LEFT.asArg()));
tap.addAction(finger.createPointerUp(PointerInput.MouseButton.LEFT.asArg()));
driver.perform(Arrays.asList(tap));
```